

90-Day SDE-2 Daily Interview Preparation Timetable

Tailored for Working Professionals (Office: 9:30 AM - 6:30 PM | Gym: 7:00 PM - 8:30 PM)

TARGET ROLE	TARGET COMPANIES	TOTAL DURATION	DAILY CAPACITY
SDE-2 / Senior Backend	Razorpay, MS, JPMC, Oracle	12 Weeks (Daily Plan)	3.5 hrs (Wkday) 8 hrs (Wknd)

Your Daily Execution Blueprint

Weekdays (Monday to Friday):	Weekends (Saturday & Sunday):
• 7:00 AM - 8:15 AM (Morning Slot 1): DSA Problem 1 (Timed, 45-60m) • 8:15 AM - 9:30 AM: Freshen up, Breakfast, Leave for Office • 9:30 AM - 6:30 PM: Office Hours (Maintain energy) • 7:00 PM - 8:30 PM: Gym Workout & Refresh • 8:30 PM - 9:30 PM: Dinner & Unwind • 9:45 PM - 10:45 PM (Night Slot 2): DSA Problem 2 (Medium) • 10:45 PM - 12:15 AM (Night Slot 3): Core Tech & Design Deep Dive	• 8:30 AM - 12:30 PM (Morning): LLD Machine Coding / HLD Design Session • 12:30 PM - 2:00 PM: Lunch & Break • 2:00 PM - 5:00 PM (Afternoon): Database Tuning / Spring Boot Internals • 5:00 PM - 6:30 PM: Walk / Rest • 7:30 PM - 10:30 PM (Evening): 3-4 Harder DSA Mediums + Mock Interviews

MONTH 1: CORE JAVA, CONCURRENCY, SOLID & FOUNDATION DSA

WEEK 01 Two Pointers, Sliding Window & Java 8–21 / Collections Internals

DAY	MORNING & NIGHT DSA PROBLEMS (EXACT NAMES)	NIGHT TECH DEEP DIVE TOPIC
Monday	AM Two Sum II - Input Array Is Sorted LC 167 PM 3Sum LC 15	Java Stream API Internals: Lazy evaluation, short-circuiting, flatMap vs map, Collectors.groupingBy() custom collectors.
Tuesday	AM Container With Most Water LC 11 PM Trapping Rain Water LC 42	Modern Java (Java 11 to 21): Records, Pattern Matching for switch, Sealed classes, and Virtual Threads (Project Loom mechanics).
Wednesday	AM Longest Substring Without Repeating Characters LC 3 PM Minimum Size Subarray Sum LC 209	HashMap Internals Under the Hood: Bucket indexing formula, collision resolution, treeification threshold (8), Load Factor 0.75 resizing.
Thursday	AM Permutation in String LC 567 PM Sliding Window Maximum LC 239	ConcurrentHashMap Mechanics: Java 7 Segment Locks vs Java 8+ CAS + synchronized bins, size calculation, tree bins.
Friday	AM Subarray Sum Equals K LC 560 PM Group Anagrams LC 49	Collections Contracts & CopyOnWrite: equals() & hashCode() rules, CopyOnWriteArrayList write-cost trade-offs.
Saturday	Wknd Longest Repeating Character Replacement LC 424 Wknd Minimum Window Substring LC 76	LLD Practice 1: Implement In-Memory LRU Cache with O(1) Get/Put and eviction policy from scratch in Java with unit tests.
Sunday	Wknd 4Sum LC 18 + Product of Array Except Self LC 238	Weekly Revision & Mock: Re-solve 3 failed problems + Deep recap of HashMap & ConcurrentHashMap source code.

WEEK 02 Binary Search, Stacks & Java Multithreading / Concurrency

DAY	MORNING & NIGHT DSA PROBLEMS	NIGHT TECH DEEP DIVE TOPIC
Monday	AM Search in Rotated Sorted Array LC 33 PM Find Minimum in Rotated Sorted Array LC 153	Java Memory Model (JMM): CPU caches, memory barriers, instruction reordering, volatile keyword visibility guarantee.
Tuesday	AM Search a 2D Matrix LC 74 PM Koko Eating Bananas LC 875	Synchronization & Monitor Locks: synchronized keyword bytecode (monitorenter/monitorexit), biased/lightweight/heavy locks.
Wednesday	AM Daily Temperatures (Monotonic Stack) LC 739 PM Next Greater Element II LC 503	Atomic Variables & Lock-Free Programming: CAS (Compare-And-Swap), AtomicInteger, AtomicReference, ABA problem.
Thursday	AM Largest Rectangle in Histogram LC 84 PM Min Stack LC 155	Explicit Locks: ReentrantLock vs synchronized, fairness policy, ReentrantReadWriteLock, StampedLock optimistic reads.
Friday	AM Evaluate Reverse Polish Notation LC 150 PM Asteroid Collision LC 735	Thread Deadlock, Livelock & Starvation: 4 deadlock conditions, reproducing a deadlock in Java, Thread Dump analysis using jstack.
Saturday	Wknd Capacity To Ship Packages Within D Days LC 1011 Wknd Median of Two Sorted Arrays LC 4	LLD Practice 2: Implement a Thread-Safe Rate Limiter (Token Bucket Algorithm) from scratch in Java with unit tests.
Sunday	Wknd Online Stock Span LC 901 + Valid Parentheses LC 20	Weekly Concurrency Review: Solve Producer-Consumer problem using wait()/notify(), BlockingQueue, and Condition.

WEEK 03 Linked Lists, Fast/Slow Pointers & java.util.concurrent (JUC)

DAY	MORNING & NIGHT DSA PROBLEMS	NIGHT TECH DEEP DIVE TOPIC
Monday	AM Reverse Linked List LC 206 PM Reverse Nodes in k-Group LC 25	ThreadPoolExecutor Deep Dive: corePoolSize, maxPoolSize, keepAliveTime, ArrayBlockingQueue vs SynchronousQueue.
Tuesday	AM Linked List Cycle II LC 142 PM Reorder List LC 143	Rejection Handlers & Thread Safety: AbortPolicy, CallerRunsPolicy, DiscardOldestPolicy, Graceful shutdown (shutdown() vs shutdownNow()).
Wednesday	AM Merge k Sorted Lists (PQ / Divide & Conquer) LC 23 PM Copy List with Random Pointer LC 138	Concurrency Utilities: CountdownLatch vs CyclicBarrier, Semaphore for resource throttling, Exchanger.
Thursday	AM LRU Cache LC 146 PM LFU Cache LC 460	CompletableFuture Asynchronous Programming: supplyAsync(), chaining with thenApply/thenCompose, allOf(), custom executor delegation.
Friday	AM Remove Nth Node From End of List LC 19 PM Add Two Numbers LC 2	ThreadLocal & Context Propagation: ThreadLocal memory leaks in pooled threads, Scoped Values in modern Java.
Saturday	Wknd Design Circular Queue LC 622 Wknd Design Browser History LC 1472	LLD Practice 3: Implement a Multi-Floor Parking Lot System with Strategy Pattern for fee calculation (Vehicle, Slot, Ticket, FeeStrategy).
Sunday	Wknd Intersection of Two Linked Lists LC 160 + Revision	Weekly Concurrency Review: Implement a custom thread pool using core Java multithreading without using Executors.

WEEK 04 Binary Trees (DFS/BFS), BST & JVM Memory / GC Mechanics

DAY	MORNING & NIGHT DSA PROBLEMS	NIGHT TECH DEEP DIVE TOPIC
Monday	AM Binary Tree Level Order Traversal LC 102 PM Binary Tree Zigzag Level Order LC 103	JVM Memory Architecture: JVM Stack frame layout, Heap space (Young Gen: Eden, S0, S1; Old Gen), Metaspace vs PermGen.
Tuesday	AM Lowest Common Ancestor of a Binary Tree LC 236 PM Diameter of Binary Tree LC 543	Garbage Collection Algorithms: Mark-Sweep-Compact, Minor GC vs Major/Full GC, Stop-the-world (STW) pauses, throughput vs latency.
Wednesday	AM Binary Tree Maximum Path Sum LC 124 PM Construct Binary Tree from Preorder and Inorder LC 105	Modern G1 GC & ZGC Internals: Region-based heap partitioning, Remembered Sets (RSet), Card Tables, ZGC colored pointers.
Thursday	AM Validate Binary Search Tree LC 98 PM Kth Smallest Element in a BST LC 230	Memory Leaks & Profiling: Unclosed resources, static collections, diagnosing OutOfMemoryError (OOM) via Eclipse Memory Analyzer (MAT).
Friday	AM Serialize and Deserialize Binary Tree LC 297 PM Populating Next Right Pointers in Each Node LC 116	JVM ClassLoader Subsystem: Bootstrap, Extension/Platform, Application ClassLoaders, Delegation model, ClassNotFoundException vs NoClassDefFoundError.
Saturday	Wknd Binary Tree Right Side View LC 199 Wknd Flatten Binary Tree to Linked List LC 114	LLD Practice 4: Implement Splitwise (Expense Sharing App) supporting Equal, Exact, and Percentage splits using Strategy & Observer patterns.
Sunday	Wknd Invert Binary Tree LC 226 + Month 1 DSA Revision	Month 1 Milestone Assessment: Full 90-minute live mock test covering Concurrency, Collections internals, and Tree algorithms.

MONTH 2: SPRING BOOT INTERNALS, POSTGRESQL OPTIMIZATION, REDIS & KAFKA

WEEK 05 Graphs (BFS/DFS, Topological Sort) & Spring Boot Core Internals

DAY	MORNING & NIGHT DSA PROBLEMS	NIGHT TECH DEEP DIVE TOPIC
Monday	AM Number of Islands LC 200 PM Max Area of Island LC 695	Spring IoC & Bean Lifecycle: BeanFactory vs ApplicationContext, Bean instantiation, BeanPostProcessor, @PostConstruct.
Tuesday	AM Clone Graph LC 133 PM Rotting Oranges LC 994	Bean Scopes & Circular Dependencies: Singleton vs Prototype thread safety, 3-level cache in Spring to resolve circular dependency.
Wednesday	AM Course Schedule (Topological Sort / Kahn's) LC 207 PM Course Schedule II LC 210	Spring AOP & Proxy Mechanics: JDK Dynamic Proxies vs CGLIB proxy bytecode enhancement, Aspect, JoinPoint, Pointcut expressions.
Thursday	AM Pacific Atlantic Water Flow LC 417 PM Surrounded Regions LC 130	The Self-Invocation Issue: Why @Transactional and @Async fail when called internally, solving with self-injection or AspectJ.
Friday	AM Number of Connected Components LC 323 PM Graph Valid Tree LC 261	Auto-Configuration Mechanics: @SpringBootApplication, @EnableAutoConfiguration, @ConditionalOnProperty, custom starter.
Saturday	Wknd Word Ladder (Bidirectional BFS) LC 127 Wknd Alien Dictionary LC 269	LLD Practice 5: Implement Movie Ticket Booking (BookMyShow) with in-memory seat concurrency lock (Pessimistic vs TTL lock).
Sunday	Wknd Accounts Merge (Union Find) LC 721 + Review	Spring Architecture Review: Code a Spring Boot custom filter, interceptor, and global @ControllerAdvice exception handler.

WEEK 06 Shortest Path, Disjoint Sets & PostgreSQL / Hibernate Optimization

DAY	MORNING & NIGHT DSA PROBLEMS	NIGHT TECH DEEP DIVE TOPIC
Monday	AM Network Delay Time (Dijkstra) LC 743 PM Cheapest Flights Within K Stops LC 787	Hibernate Entity Lifecycles: Transient, Managed (Persistent), Detached, Removed states; First-level Session cache and dirty check.
Tuesday	AM Redundant Connection LC 684 PM Is Graph Bipartite? LC 785	The N+1 Query Problem: Lazy vs Eager fetch trap; resolving via JOIN FETCH, @EntityGraph, and default_batch_fetch_size.
Wednesday	AM Swim in Rising Water LC 778 PM Path with Minimum Effort LC 1631	Transaction Isolation Levels & Phenomena: Read Committed vs Repeatable Read vs Serializable; Dirty reads, Phantom reads.
Thursday	AM Min Cost to Connect All Points (Prim's/Kruskal's) LC 1584 PM Reconstruct Itinerary LC 332	Concurrency & Database Locking: Optimistic Locking (@Version) vs Pessimistic Locking (SELECT FOR UPDATE / PESSIMISTIC_WRITE).
Friday	AM Find Eventual Safe States LC 802 PM Critical Connections in a Network (Tarjan's) LC 1192	Postgres Indexing & Query Tuning: B-Tree mechanics, Composite indexes column order, Covered index, analyzing EXPLAIN ANALYZE.
Saturday	Wknd Word Search II (Trie + Backtracking) LC 212 Wknd Design Add and Search Words Data Structure LC 211	PostgreSQL Deep Dive: Table partitioning (range/hash), connection pooling (HikariCP parameters), and transaction propagation rules.
Sunday	Wknd Implement Trie (Prefix Tree) LC 208 + Review	System Design 1: Design a Scalable URL Shortener (TinyURL) - DB Schema, base62 encoding, Redis caching, and collision handling.

WEEK 07 Heaps, Priority Queues & Distributed Caching (Redis)

DAY	MORNING & NIGHT DSA PROBLEMS	NIGHT TECH DEEP DIVE TOPIC
Monday	AM Kth Largest Element in an Array LC 215 PM Top K Frequent Elements LC 347	Redis Core Architecture: In-memory single-threaded event loop, Data structures (Strings, Hashes, Lists, Sets, Sorted Sets / ZSets).
Tuesday	AM Find Median from Data Stream LC 295 PM Task Scheduler LC 621	Caching Strategies: Cache-Aside, Read-Through, Write-Through, Write-Back; Eviction policies (LRU, LFU, volatile-ttl).
Wednesday	AM K Closest Points to Origin LC 973 PM Reorganize String LC 767	Cache Pitfalls & Mitigation: Cache Avalanche (jittered TTL), Cache Penetration (Bloom Filter), Cache Stampede / Breakdown (Mutex).
Thursday	AM Last Stone Weight LC 1046 PM Seat Reservation Manager LC 1845	Distributed Locking with Redis: SETNX with TTL, Redlock algorithm, lock renewal (Watchdog pattern), handling split-brain.
Friday	AM Design Twitter (Heap + OOP) LC 355 PM Single-Threaded CPU LC 1834	Redis High Availability: Master-Slave replication, Redis Sentinel for automatic failover, Redis Cluster (16384 hash slots).
Saturday	Wknd Find K Pairs with Smallest Sums LC 373 Wknd IPO (Max Capital) LC 502	System Design 2: Design a Real-Time Gaming Leaderboard / Stock Ticker using Redis Sorted Sets (ZADD, ZREVRANGEBYSCORE).
Sunday	Wknd Sort Characters By Frequency LC 451 + Review	Hands-on Redis: Implement a Redis Cache-Aside layer with Spring Boot @Cacheable and custom RedisTemplate serialization.

WEEK 08 Dynamic Programming Foundations & Kafka Event Streaming

DAY	MORNING & NIGHT DSA PROBLEMS	NIGHT TECH DEEP DIVE TOPIC
Monday	AM Climbing Stairs LC 70 PM Min Cost Climbing Stairs LC 746	Kafka Architecture: Topics, Partitions, Segment files, Producers, Consumers, Consumer Groups, ZooKeeper vs KRaft mode.
Tuesday	AM House Robber LC 198 PM House Robber II LC 213	Partitioning & Message Ordering: Key-based hash partitioning, strict partition ordering, consumer group rebalancing protocol.
Wednesday	AM Coin Change LC 322 PM Coin Change II (Combinations) LC 518	Delivery Semantics & Offsets: At-most-once, At-least-once, Exactly-once (Kafka Transactions), Auto-commit vs Manual commit.
Thursday	AM Longest Increasing Subsequence LC 300 PM Maximum Subarray (Kadane's) LC 53	Handling Failures in Kafka: Poison pill messages, Dead Letter Queues (DLQ), retry topics with exponential backoff, consumer lag monitoring.
Friday	AM Partition Equal Subset Sum LC 416 PM Target Sum LC 494	Resilience4j Microservices Patterns: CircuitBreaker state transitions (CLOSED/OPEN/HALF_OPEN), RateLimiter, Bulkhead, Fallback.
Saturday	Wknd Decode Ways LC 91 Wknd Word Break LC 139	System Design 3: Design a High-Throughput Notification Service (Email/SMS/Push) using Kafka, Redis rate limiters, and idempotency store.
Sunday	Wknd Combination Sum IV LC 377 + Month 2 Review	Month 2 Milestone Assessment: Full 90-minute live coding test on Spring Boot, PostgreSQL locking, and Kafka architecture.

MONTH 3: FINTECH SYSTEMS, REACT INTEGRATION, RESUME & MOCKS

WEEK 09 2D Dynamic Programming & FinTech / Distributed Transactions

DAY	MORNING & NIGHT DSA PROBLEMS	NIGHT TECH DEEP DIVE TOPIC
Monday	AM Unique Paths LC 62 PM Unique Paths II (Obstacles) LC 63	Idempotency in Distributed Systems: Unique Idempotency-Key headers, Redis/DB deduplication lock, replay attack mitigation.
Tuesday	AM Longest Common Subsequence LC 1143 PM Edit Distance LC 72	Distributed Saga Pattern: Choreography-based Saga vs Orchestration-based Saga, Compensating transactions during payment failures.
Wednesday	AM Best Time to Buy and Sell Stock with Cooldown LC 309 PM Coin Change (Space Optimized)	Transactional Outbox Pattern: Solving dual-write problems between PostgreSQL database and Apache Kafka without distributed 2PC.
Thursday	AM Maximum Product Subarray LC 152 PM Palindromic Substrings LC 647	Double-Entry Ledger Design: Immutable credit/debit transaction records, preventing rounding errors, high-scale balance calculation.
Friday	AM Longest Palindromic Substring LC 5 PM Interleaving String LC 97	Webhook Delivery System: Reliable webhook dispatch, exponential backoff retries, signature verification (HMAC SHA-256).
Saturday	Wknd Burst Balloons LC 312 Wknd Regular Expression Matching LC 10	System Design 4 (Razorpay Tier): Design a Payment Gateway Processing Engine & Webhook Service with Idempotency & Saga.
Sunday	Wknd Distinct Subsequences LC 115 + Review	FinTech Architecture Review: Write working Java code for an Idempotency Interceptor checking Redis before executing service methods.

WEEK 10 Intervals, Backtracking & React Full-Stack Architecture		
DAY	MORNING & NIGHT DSA PROBLEMS	NIGHT TECH DEEP DIVE TOPIC
Monday	AM Merge Intervals LC 56 PM Insert Interval LC 57	React Core Internals: Virtual DOM reconciliation (Fiber architecture), component re-rendering triggers, key prop significance.
Tuesday	AM Non-overlapping Intervals LC 435 PM Meeting Rooms II LC 253	React Hooks Performance: useEffect dependency array pitfalls, useMemo, useCallback, memoizing child components.
Wednesday	AM Subsets LC 78 PM Subsets II (Duplicates) LC 90	State Management Architectures: Prop drilling vs React Context vs Redux Toolkit / Zustand; Client State vs Server State (React Query).
Thursday	AM Permutations LC 46 PM Combination Sum LC 39	Full-Stack Security & Auth: JWT storage (HttpOnly SameSite Cookies vs LocalStorage XSS risks), CSRF tokens, Axios Interceptors.
Friday	AM Letter Combinations of a Phone Number LC 17 PM Generate Parentheses LC 22	CORS & API Gateway Integration: Pre-flight OPTIONS request handling, CORS headers (Access-Control-Allow-Origin) on Spring Cloud Gateway.
Saturday	Wknd N-Queens LC 51 Wknd Sudoku Solver LC 37	System Design 5: Design a Flash Sale / E-Commerce Inventory Lock System (High-concurrency Redis decrement + DB write-back).
Sunday	Wknd Palindrome Partitioning LC 131 + Review	Full-Stack Architecture Review: Diagram end-to-end data flow from React client → API Gateway → Microservices → Postgres.

WEEK 11 DSA Weak Area Drills & Resume STAR Refactoring		
DAY	MORNING & NIGHT DSA DRILLS	NIGHT DELIVERABLE / PREPARATION
Monday	AM Re-solve 2 previously failed Sliding Window Mediums PM Re-solve 1 Hard Monotonic Stack problem	Resume Overhaul 1: Refactor Project 1 into STAR format: Problem → Architectural Solution → Quantified Metric (RPS/Latency).
Tuesday	AM Re-solve 2 Tree LCA / Serialization Mediums PM Re-solve 1 BST Validation problem	Resume Overhaul 2: Refactor Project 2 highlighting Microservices, Docker, Kafka event-driven pipelines, and DB index tuning.
Wednesday	AM Re-solve 2 Graph Topological Sort Mediums PM Re-solve 1 Dijkstra Shortest Path problem	LinkedIn & Portal Optimization: Update LinkedIn headline to "SDE-2 / Senior Backend Engineer Java, Spring Boot, Distributed Systems".
Thursday	AM Re-solve 2 Dynamic Programming Coin Change/LIS PM Re-solve 1 Heap Top-K problem	Referral Strategy: Craft crisp 3-sentence outreach templates for Senior Engineers and Tech Leads on LinkedIn.
Friday	AM LRU Cache (Time trial: 20 mins blank IDE) PM Rate Limiter (Time trial: 25 mins blank IDE)	Behavioral / Leadership Rounds: Prepare 5 core STAR stories: Technical conflict, Production outage, Difficult deadline, Architecture win.
Saturday	Wknd Full Mock Interview 1 (DSA): 2 unseen LeetCode Medium problems in 45 mins with think-aloud practice.	Full Mock Interview 2 (LLD Machine Coding): Build complete Splitwise or Parking Lot in 90 mins with clean design.
Sunday	Wknd Full Mock Interview 3 (System Design HLD): 45-minute live whiteboard simulation of Razorpay Payment Gateway.	Pipeline Launch: Submit first batch of 15 targeted referral applications on LinkedIn, Instahyre, and career portals.

WEEK 12 Aggressive Interview Execution & Continuous Revision

DAY	DAILY ROUTINE & WARM-UP DSA	LIVE PIPELINE & OFFER CONVERSION
Monday	AM 1 Quick Warm-Up DSA Medium (Array/Window)	Send 5 targeted LinkedIn referral messages + Follow-up on active applications on Instahyre.
Tuesday	AM 1 Quick Warm-Up DSA Medium (Tree/Graph)	Review company-specific past interview experiences on LeetCode Discuss for scheduled interview rounds.
Wednesday	AM 1 Quick Warm-Up DSA Medium (Binary Search)	Quick revision: Java Concurrency memory barriers, ConcurrentHashMap source code, and G1 GC pauses.
Thursday	AM 1 Quick Warm-Up DSA Medium (Monotonic Stack)	Quick revision: PostgreSQL indexing strategies, N+1 query mitigations, and isolation level locks.
Friday	AM 1 Quick Warm-Up DSA Medium (DP / Intervals)	Quick revision: Kafka partition balancing, DLQs, Redis distributed locking, and Saga pattern.
Saturday	Wknd Active Interview Weekend: Attend scheduled company coding rounds / machine coding rounds.	Post-Interview Retrospective: Log every question asked, review any gaps, and refine answers for subsequent rounds.
Sunday	Wknd Weekly Review & Pipeline Refill: Apply to next batch of 15 product engineering openings.	Continuous Momentum: Maintain daily 2-problem habit until you receive and sign your target product offer.